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week 11 activity 01

1.

$$C(s) = \frac{K(1+T_L s)}{(1+\alpha T_L s)}$$

1.1 $$C(j\omega) = \frac{K(1+T_L j\omega)}{(1+\alpha T_L j\omega)}$$

$$= \frac{K(1+jT_L\omega)}{(1+j\alpha T_L\omega)} = \frac{K(1+jT_L\omega)(1-j\alpha T_L\omega)}{1+\alpha^2 T_L^2\omega^2} = \cancel{K[(1+\alpha T_L^2\omega^2)+}$$

$$|C(j\omega)| = \frac{K\sqrt{1+(T_L\omega)^2}}{\sqrt{(1+\alpha^2 T_L^2\omega^2)}} = \frac{K[(1+\alpha T_L^2\omega^2)+j\omega T_L(1-\alpha)]}{1+\alpha^2 T_L^2\omega^2}$$

$$\cancel{\tan^{-1}} \quad \angle C(j\omega) = \cancel{\tan^{-1}} 1$$

$$\angle C(j\omega) = \tan^{-1}\left[\frac{\omega T_L(1-\alpha)}{(1+\alpha T_L^2\omega^2)}\right]$$

$\omega = 0$ $|C(0)| = K$ $\angle C(0) = 0$

$\omega = \infty$ $|C(\infty)| = \frac{K}{\alpha}$ $\angle C(\infty) = 0$

$Im(C(j\omega)) = 0$, $\frac{\omega T_L(1-\alpha)}{1+\alpha^2 T_L^2\omega^2} = 0$, $\omega = 0$

$Re(C(j\omega)) = 0$ $\frac{1+\alpha T_L^2\omega^2}{1+\alpha^2 T_L^2\omega^2} = 0$ ω does not cross real axis.

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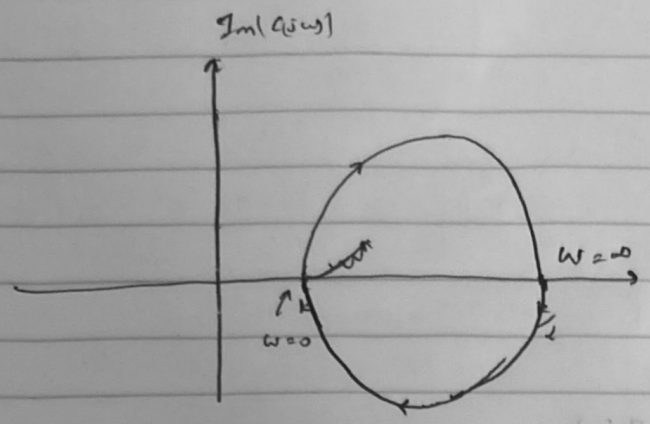

1.2.

$$\frac{d\,\mathrm{Im}(G(j\omega))}{d\omega} = \frac{T_L(1-\alpha)(1+\omega^2T_L^2\alpha) - 2\omega^2T_L(1-\alpha)T_L^2\alpha}{(1+\omega^2T_L^2\alpha)^2}$$

$$= \frac{T_L(1+\omega^2T_L^2\alpha - \alpha - \omega^2T_L^2\alpha^2 - 2\omega^2T_L^2\alpha + 2\omega^2T_L^2\alpha^2)}{(1+\omega^2T_L^2\alpha^2)^2}$$

$$= \frac{T_L(1 - \omega^2T_L^2\alpha - \alpha + \omega^2T_L^2\alpha^2)}{(1+\omega^2T_L^2\alpha^2)^2}$$

$$= \frac{T_L((1-\alpha) - \omega^2T_L^2\alpha(1-\alpha))}{(1+\omega^2T_L^2\alpha^2)^2}$$

$$= \frac{T_L((1-\alpha)(1-\omega^2T_L^2\alpha))}{(1+\omega^2T_L^2\alpha^2)^2}$$

$$1-\omega^2T_L^2\alpha = 0 \quad , \quad \frac{d\,\mathrm{Im}(G(j\omega))}{d\omega} = 0.$$

$$W = \frac{1}{T_L\sqrt{\alpha}} \quad \max \quad \mathrm{Im}(G(j\omega)).$$

$$\mathrm{Im}(G(j\omega))_{max} = \frac{1-\alpha}{1+\alpha}$$

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Therefore maximum ϕ_m $\sin\phi_m = \left(\frac{1-\alpha}{1+\alpha}\right)$

e.3 $C(j\omega) = \frac{K(1+jT_L\omega)}{1+jT_L\omega\alpha}$

$|C(j\omega)| = K\sqrt{\frac{(1+\omega^2T_L^2)}{(1+\alpha^2\omega^2T_L^2)}}$

$|C(j\omega)|_{dB} = 20\log \frac{K\sqrt{(1+\omega^2T_L^2)}}{(1+\alpha^2\omega^2T_L^2)^2}$

$= 20\log K^2(1+\omega^2T_L^2) - 20\log(1+\alpha^2\omega^2T_L^2)$

$\angle C(j\omega) = \tan^{-1}(T_L\omega) - \tan^{-1}(T_L(\omega)\alpha)$

~~$\omega \to 0$~~

$0 < \omega < \frac{1}{T_L}$ $|C(j\omega)|_{dB} = 20\log K^2$

$\frac{1}{T_L} < \omega < \frac{1}{\alpha T_L}$ $|C(j\omega)|_{dB} \to 20$ db/decade.

$\frac{1}{\alpha T_L} < \omega$ $|C(j\omega)|_{dB} \to 0$ db/decade.

ω small $\angle C(j\omega) \to 0$

ω large $\angle C(j\omega) \to 0$

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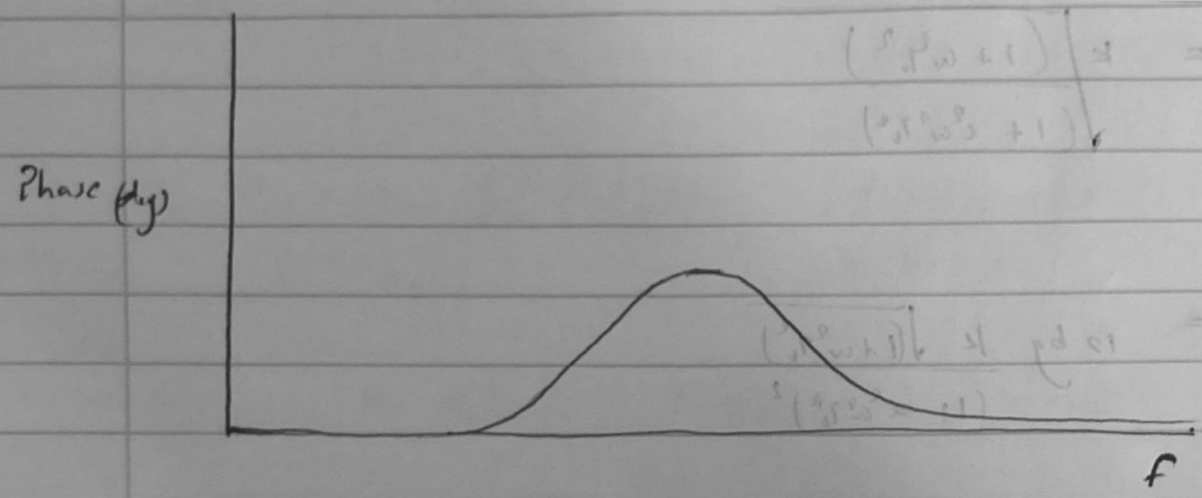
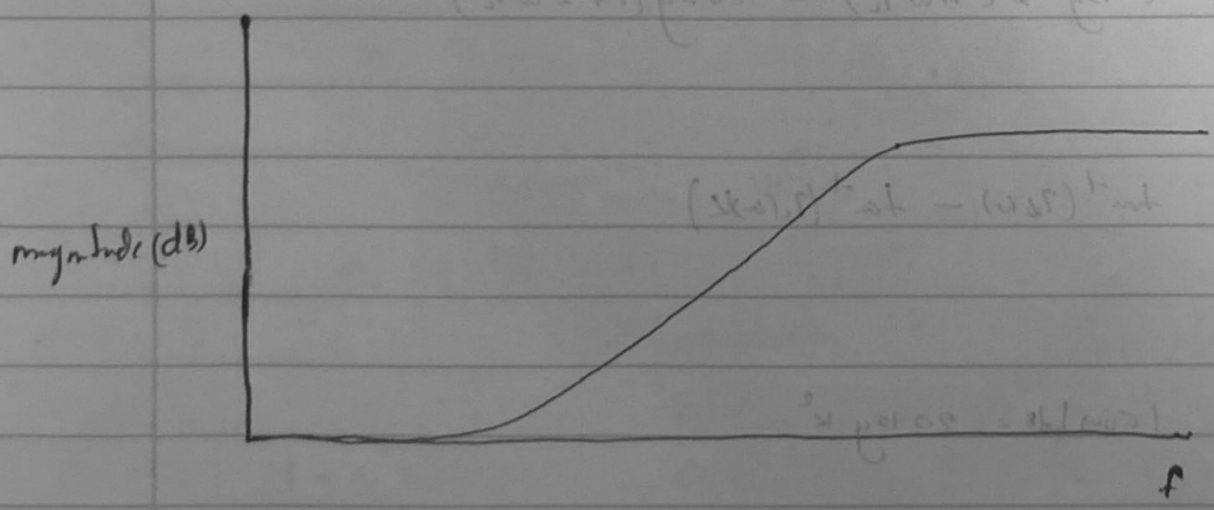

Phase (deg)
f


magnitude (dB)
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1.4) $G(s) = \dfrac{2(s+20)}{(s+1)(s+2)(s+3)}$

from Bode plot

Phase margin = ~~19.8°~~ 20°

gain crossover frequency = ~~2.813 rad/s~~ 2.75 rad/s

gain margin = 6dB

phase crossover frequency = 4 rad/s.

1.5) required increase in phase margin = 50 − 20
= 30°

$\omega_m = \dfrac{1}{T_L\sqrt{\alpha}}$, $\sin\phi_m = \dfrac{1-\alpha}{1+\alpha}$.

$\dfrac{1-\alpha}{1+\alpha} = \sin 30$, $T_L = \dfrac{1}{\omega_m\sqrt{\alpha}}$

$2 - 2\alpha = 1+\alpha$ $= \dfrac{1}{2.75 \times \frac{1}{\sqrt{3}}}$

$\alpha = \dfrac{1}{3}$. $= 0.63$ s

∴ required ~~[illegible]~~ transfer $C(s) = \dfrac{K(1+0.63s)}{1+0.21s}$

Atlas

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